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Laboratory I
Optical/X-ray Study of AGN

Abstract

The goal of this exercise is to analyse the optical and X-ray spectral data from an Active Galactic Nucleus (AGN) to determine its physical properties, such as redshift, the widths and strengths of the main emission lines, and X-ray spectral parameters. The techniques used consist of spectral fitting for both optical and X-ray data, and Bayesian analysis to extract relevant information from the X-ray spectrum. The main results include the redshift of the AGN, the velocity dispersion derived from the optical emission lines, and the best-fit parameters for the X-ray spectral models, which indicate significant absorption, which indicates that the AGN is likely a Type 2 AGN. The combination of optical and X-ray analyses provides a comprehensive understanding of the AGN's characteristics and supports the AGN Unified Model.

1 Methods

All the methods used in this exercise are based on those explained in the lab guide and in the tutorials provided, mainly the Peter Boorman tutorial [1] and the Johannes Buchner Xspec Bayesian analysis guide [2]. These are carried out using Python programming language [3], the Pyxspec library [4] for X-ray data analysis, which is not easy to install nor use but a very powerful tool. The Astropy library [5] is also used to model and fit the optical spectrum.

The optical spectrum analysis involves reading and plotting the spectral data from FITS files, identifying key emission and absorption lines, which allows to determine the redshift of the AGN, and fitting Gaussian profiles to the emission lines to derive their Full Width at Half Maximum (FWHM) and velocity dispersion.

Before fitting the lines, the spectrum is shifted to the rest frame using the first approximation of the redshift obtained from the pipeline data with the following code from the link provided in the lab guide [6]:

```
spectra_pipeline_data = Table.read('spec-2.fits', format='fits',hdu=2)
spectra_z=spectra_pipeline_data[0]['Z']
spectra_z_err=spectra_pipeline_data[0]['Z_ERR']
```

With a similar process, some important data can be read from the FITS header such as the SDSS identifier and the coordinates of the AGN, which will be presented later.

To determine the redshift more accurately, several emission and absorption lines are identified in the optical spectrum. The center of these lines is measured, and the redshift is calculated for each line using the following equation:

$$z = \frac{\lambda_{obs} - \lambda_{theo}}{\lambda_{theo}} \tag{1}$$

where λ_{obs} is the wavelength observed in the spectrum, and λ_{theo} is the theoretical rest-frame wavelength of the identified emission or absorption line, ennumerated in the Tabla 1 of the lab guide.

The widths of the lines are also measured (FWHM) fitting Gaussians to the emission lines in the optical spectrum. This allows to determine the velocity dispersion of the gas emitting these lines through the Doppler broadening using the next equation:

$$\sigma_v = \frac{\mathrm{FWHM}}{2\sqrt{2\ln(2)}} \approx 0.4247 \times \mathrm{FWHM} \tag{2}$$

where FWHM is given in km/s.

Finally for the optical spectrum, the SMBH mass can be estimated using the velocity dispersion obtained from the Hβ and [MgII] emission lines, applying the empirical relations provided in the lab guide:

$$M_{BH}(H_\beta) = 1.05 \times 10^8 \left(\frac{L_{5100}}{10^{46}\ \mathrm{erg/s}}\right)^{0.65} \left[\frac{FWHM(H_\beta)}{10^3\ \mathrm{km/s}}\right]^2 M_\odot, \tag{3}$$

$$M_{BH}(MgII) = 5.6 \times 10^6 \left(\frac{L_{3000}}{10^{44}\ \mathrm{erg/s}}\right)^{0.62} \left[\frac{FWHM(MgII)}{10^3\ \mathrm{km/s}}\right]^2 M_\odot, \tag{4}$$

where L_{5100} and L_{3000} are the are the accretion disk luminosities at rest-frame 5100Å and 3000Å.

For the X-ray spectra the data reading and plotting is carried out using Pyxspec and XQuartz [7] for visualization. Following Peter Boorman's tutorial [1], up to five different models are fitted to the X-ray data: a simple power-law, an absorbed power-law, an absorbed power-law with a Gaussian component, an absorbed power-law with a Gaussian component and Compton scattering, and a double absorbed power-law. From all of these models uniform priors are assumed for all the parameters with initial values based on typical AGN properties. These parameters are finally fitted for every model using Bayesian statistics, although errors are not calculated in this exercise due to time constraints. The logZ evidence criteria is used to compare the models and determine the best-fit one altogether with an AIC analysis.

2 Results

The results obtained from the analysis of the optical and X-ray spectra of the studied AGN are presented in two main subsections.

2.1 Optical spectra

The optical spectrum of the studied AGN is analysed in this section as per the lab guide instructions. The data used corresponds to the FITS file provided in the lab guide, and the analysis is carried out as described in the Methods section. The first step is to read and plot the optical spectrum data, although this plot is shown a bit later after shifting to rest frame and identifying the lines.

2.1.1 Determination of redshift

As mentioned in the Methods section, some important data can be read from the FITS header, shown in Tab. 1:

SDSS ID	Plate	MJD	Fiber
	4200	55499	994
Coordenades	RA	Dec	
	333.165616	0.002465	

Table 1: SDSS identifier and coordinates of the studied AGN.

Again, as specified in the Methods section, a first approximation of the redshift is obtained from the pipeline data using the shown code. The value obtained can be used to calculate different distances to the AGN using a Cosmology Calculator by Ned Wright [8]. The redshift can also be calculated more accurately by identifying several emission and absorption lines in the optical spectrum, and calculating the mean redshift from these lines. Then, the distances are recalculated based on this more accurate redshift. All these values are also shown in Tab. 2:

The angular size distance, D_A, luminosity distance, D_L, and the comoving radial distance or proper distance are obtained for a flat ΛCDM universe ($\Omega_m = 0.286$, $\Omega_\Lambda = 0.714$, $H_0 = 69.6$ km s^{-1} Mpc^{-1}). The values obtained from both methods are very similar, indicating that the initial approximation from the pipeline is quite accurate. This is a very useful tool to do a preliminary analysis of the optical spectrum.

	z	D_A (Mpc)	D_L (Mpc)	Proper distance (Mpc)
First approximation	0.33266 ± 0.00003	994.4	1766.1	1325.3
Mean from lines	0.3328 ± 0.0005	994.8	1767.3	1325.9

Table 2: Redshift and distances of the studied AGN, first with an approximation from the pipeline and then the mean from identified lines.

2.1.2 Spectral fit

The spectrum in rest frame (redshift corrected) is plotted in Fig. 1 with the identified emission and absorption lines marked with their own labels and wavelengths, including the sky lines contained in the Tabla 1 of the lab guide.

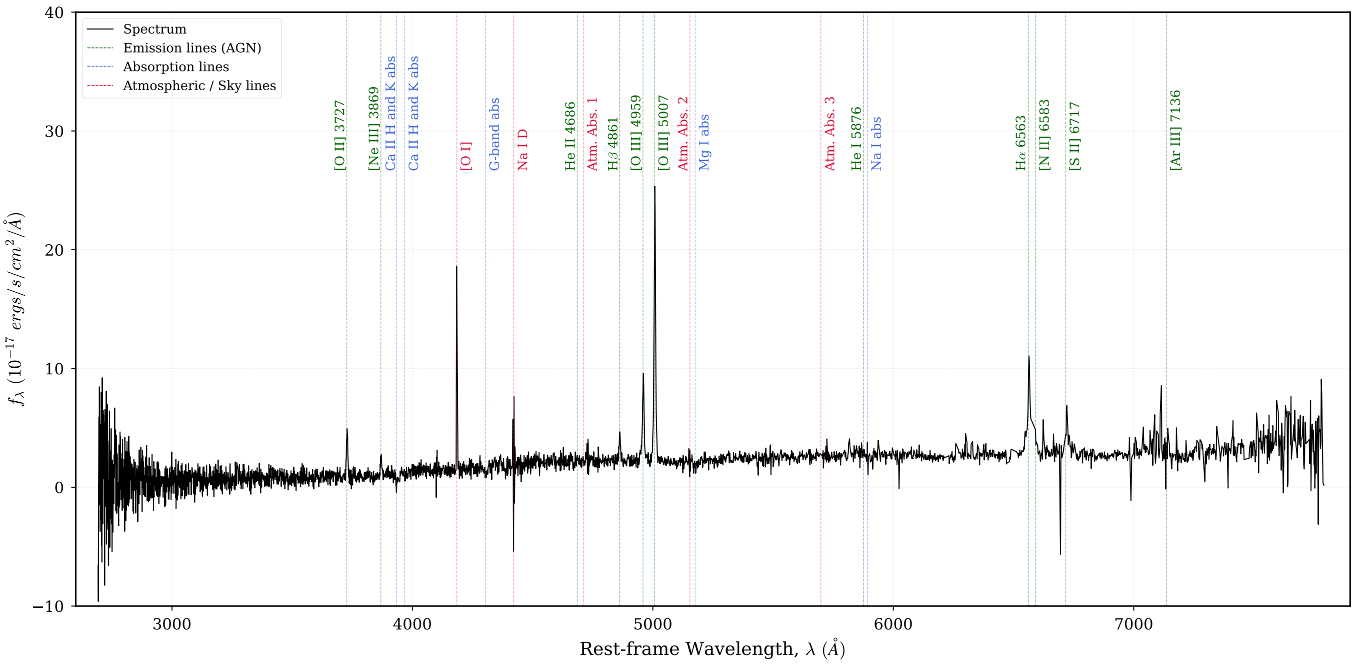

Figure 1: Optical spectrum of the studied AGN in rest frame. The identified emission and absorption lines are marked with vertical dashed lines and labels.

In the spectrum of Fig. 1, several emission lines typical of AGN can be identified in the legend of the plot. These appear to be narrow emission lines, suggesting that the AGN is likely a Type 2 AGN. For a deeper analysis of the spectrum, other significant features can be identified. On one hand, some absorption lines can be seen in the spectrum that are not marked because they are not included in the Tabla 1 of the lab guide. On the other hand, it can be appreciated that some of these included in the Tabla 1 are not detected in the spectrum.

The emission lines at low wavelengths such as Lyα 1216 Å, or and C IV 1549 Å are not detected in the spectrum. This is due to the fact that the spectrum only covers the wavelength range from approximately 2800 Å to 7800 Å in the rest-frame, so these lines fall outside this range. Additionally, the Mg II 2798 Å line is not clearly detected in the spectrum.

Some of the main emission lines identified in the optical spectrum are fitted with Gaussian profiles to determine their FWHM, which allows to calculate the velocity dispersion of the gas emitting these lines with Eq. (2). The results are shown in the following table:

Emission Line	λ_{theo} (Å)	λ_{obs} (Å)	FWHM (km/s)	σ_v (km/s)
[O II]	3727.1	3728.4 ± 0.15	6.6 ± 0.4	530 ± 30
[Ne III]	3868.8	3869.9 ± 0.4	7.1 ± 0.9	550 ± 70
Hβ	4861.3	4862.5 ± 0.8	5.5 ± 1.9	338 ± 120
[O III]	5006.8	5008.3 ± 0.1	7.1 ± 0.3	425 ± 16
Hα	6562.8	6568.3 ± 1.7	37 ± 4	1700 ± 200
[N II]	6583.5	6568.2 ± 1.8	36 ± 5	1600 ± 200
[S II]	6716.4	6724.7 ± 3.0	20 ± 7	900 ± 300

Table 3: Theoretical, λ_{theo}, and observed, λ_{obs}, wavelength, FWHM and σ_v of the main emission lines in the optical spectrum of the studied AGN.

From the results seen in Tab. 3, it can be determined that the Hβ line is narrow, according with Fig. 1. This suggests that the AGN is likely a Type 2 AGN. The limit between narrow and broad lines is typically set at $\sigma_v \approx 1000$ km/s, so the Hβ line is clearly narrow and the mass of the central SMBH cannot be estimated using this method. The median redshift calculated from these lines is previously shown in Tab. 2.

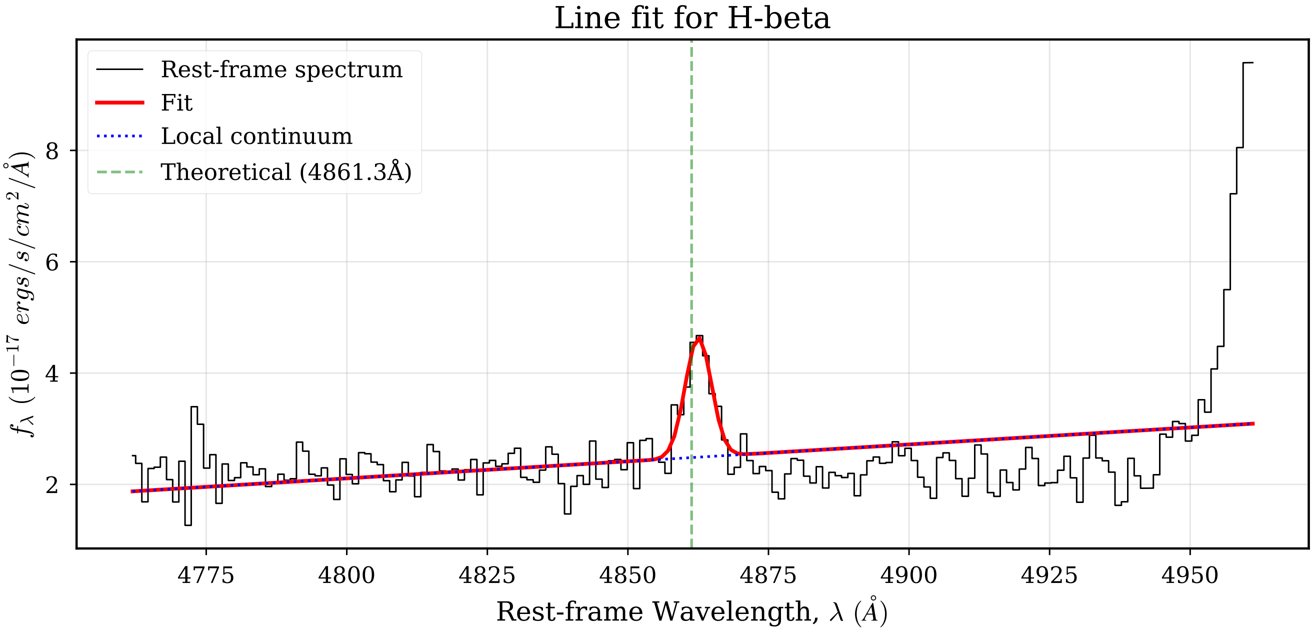

Figure 2: Spectral fit of the Hβ emission line. The rest frame spectrum is shown in black, the local continuum is shown in a dotted blue line, the fitted Gaussian profile is represented by the red curve and the theoretical wavelength is marked with a vertical dashed green line.

In Fig. 2 it can be seen that the fit is quite good, with the Gaussian profile accurately representing the shape of the Hβ emission line. The local continuum is also well estimated, and the theoretical wavelength of the Hβ line is marked for reference, although the observed wavelength is slightly shifted to the red. This is due to the fact that the first approximation of the redshift used to shift the spectrum to rest frame is a bit underestimated, as seen in Tab. 2. Then the correct redshift to use would be the mean from the lines, $z = 0.3328 \pm 0.0005$.

2.1.3 Black Hole Masses

As seen in the optical spectrum analysed, the Hβ line is narrow and the Mg II 2798 Å line is not clearly detected in the spectrum, indicating that the AGN is likely a Type 2 AGN. Therefore, it is not possible to estimate the mass of the central SMBH using Eqs. (3) and (4). In Type 2 AGN, the broad line region (BLR) is obscured by the torus, preventing the detection of broad emission lines in the optical spectrum.

2.2 X-ray spectra

The X-ray spectrum of the studied AGN is analysed in this section as per the Peter Boorman tutorial [1]. The data used corresponds to the Source 2 provided in the lab guide, and the analysis is carried out as described in the Methods section.

2.2.1 Inspect and plot the data

The X-ray data is inspected and plotted before fitting any models:

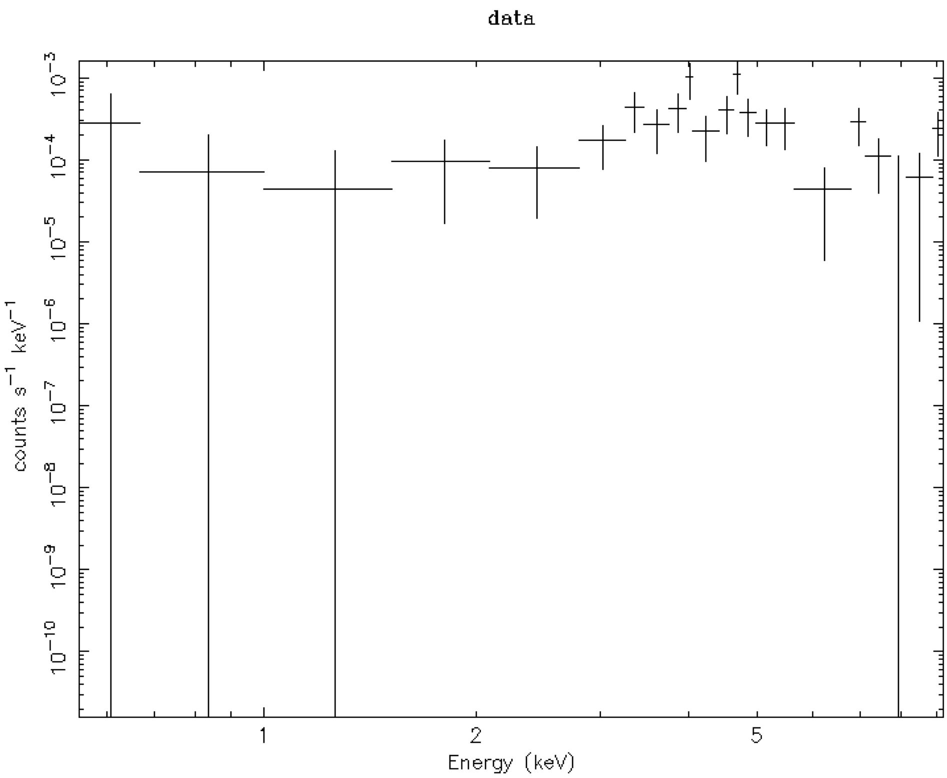

Figure 3: X-ray spectrum of the studied AGN. The counts per second are plotted as a function of energy in keV in a log-log scale.

A first inspection of the X-ray spectrum in Fig. 3 shows that a simple power-law model would not fit the data well, indicating the presence of absorption and other spectral features. This is later confirmed with the fitted models and their statistical analysis.

2.2.2 Fit the data

To fit the data, five models are used: a power-law, an absorbed power-law, an absorbed power-law with a Gaussian component, an absorbed power-law with a Gaussian component and Compton scattering and doble absorbed power-law. All these are fitted using Bayesian statistics as explained in the Methods section. A plot of the xray spectrum and a corner plot for each model is carried out, and the run and trace plots are not shown here for brevity, but can be found in Appendix A.

The first model is the simple power-law and the fit is shown in the following figure:

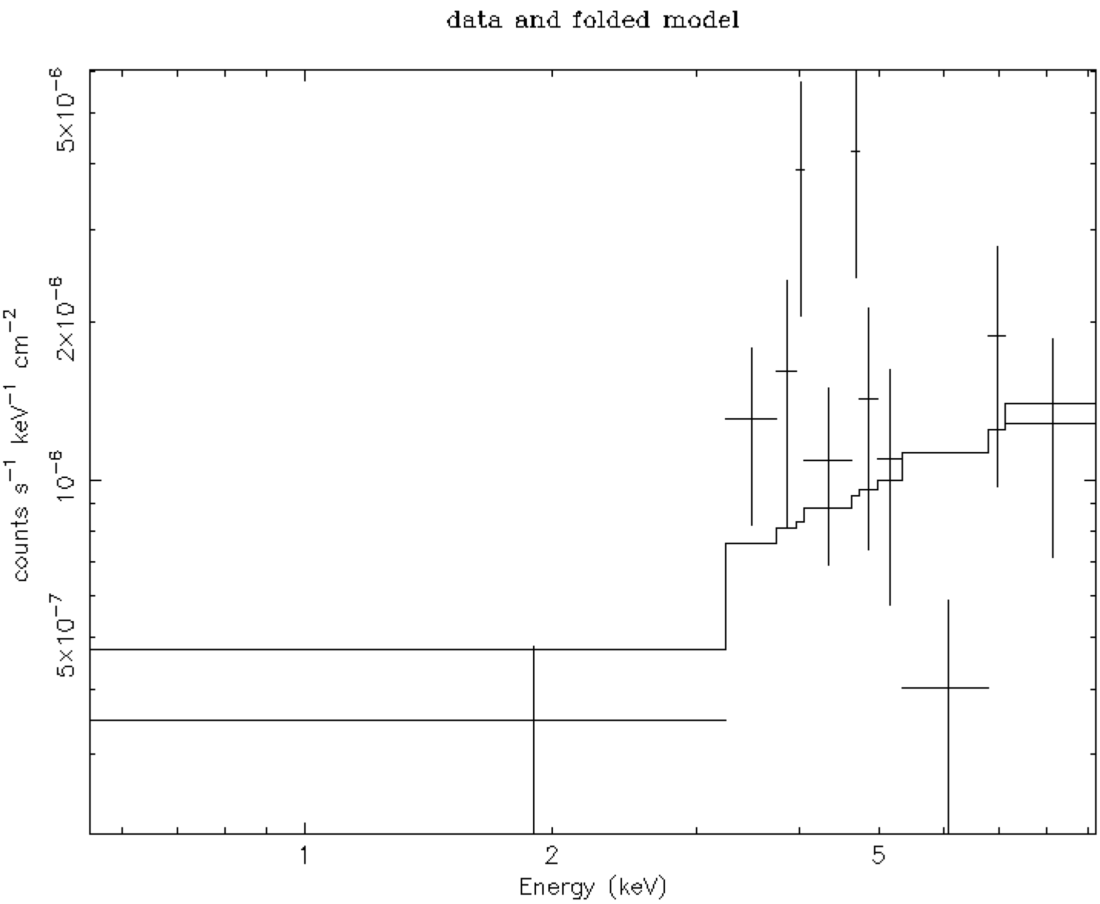

Figure 4: X-ray spectrum of the studied AGN fitted with a simple power-law model. The data points are shown in black, while the best-fit model is represented by the line.

Where it can be seen that the model does not fit well the data at all. The low-energy part of the spectrum is not well represented, indicating significant absorption that is not accounted for in this simple model.

The corner plot for the simple power-law model is shown in the next figure:

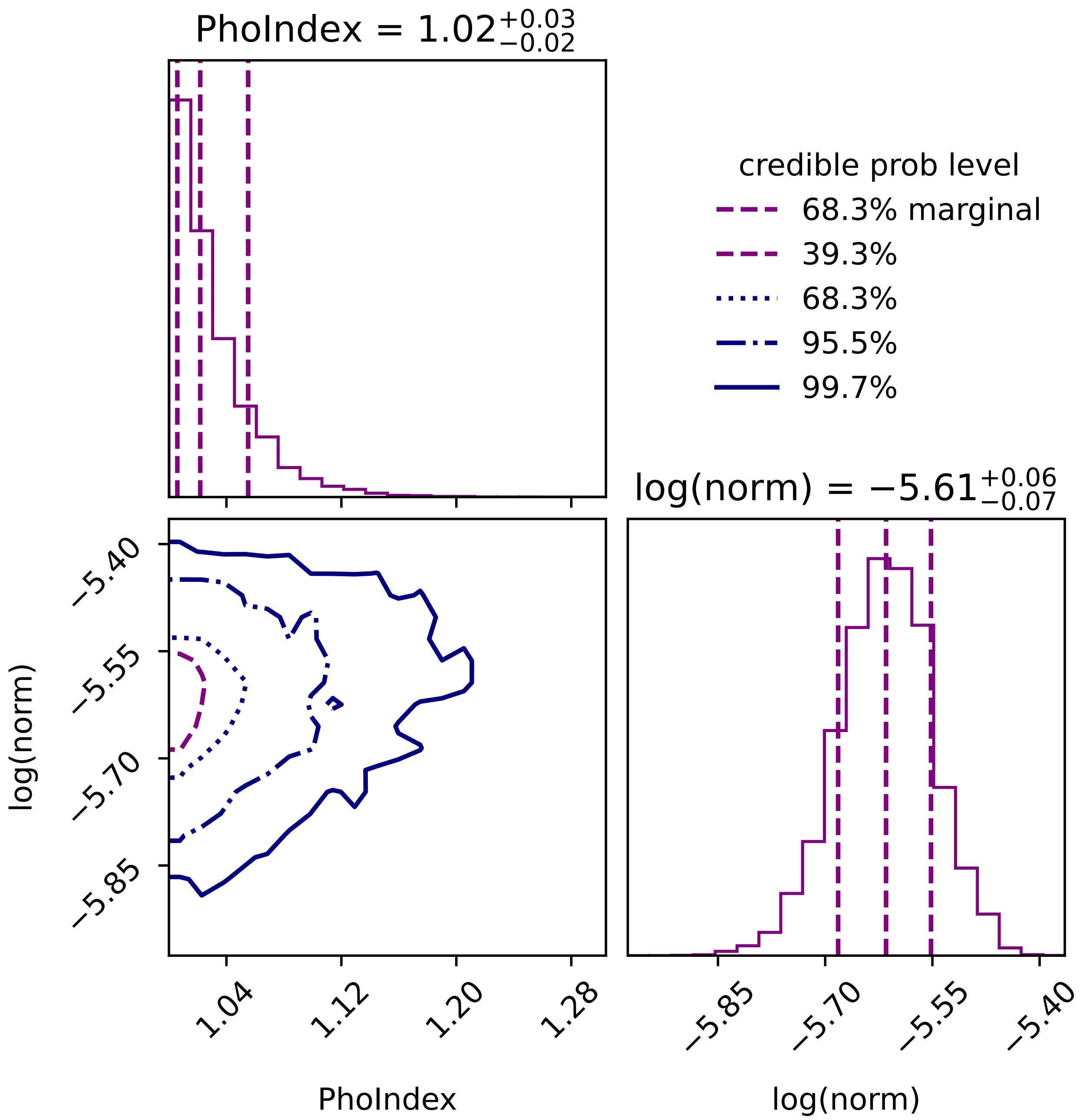

Figure 5: Corner plot of the fitted parameters for the simple power-law model. The histograms along the diagonal represent the posterior distributions of each parameter, the photon index and the normalization, while the off-diagonal plot show the joint distributions between this pair of parameters, indicating correlations and covariances.

The second model is the absorbed power-law whose fit is shown in Fig. 6.

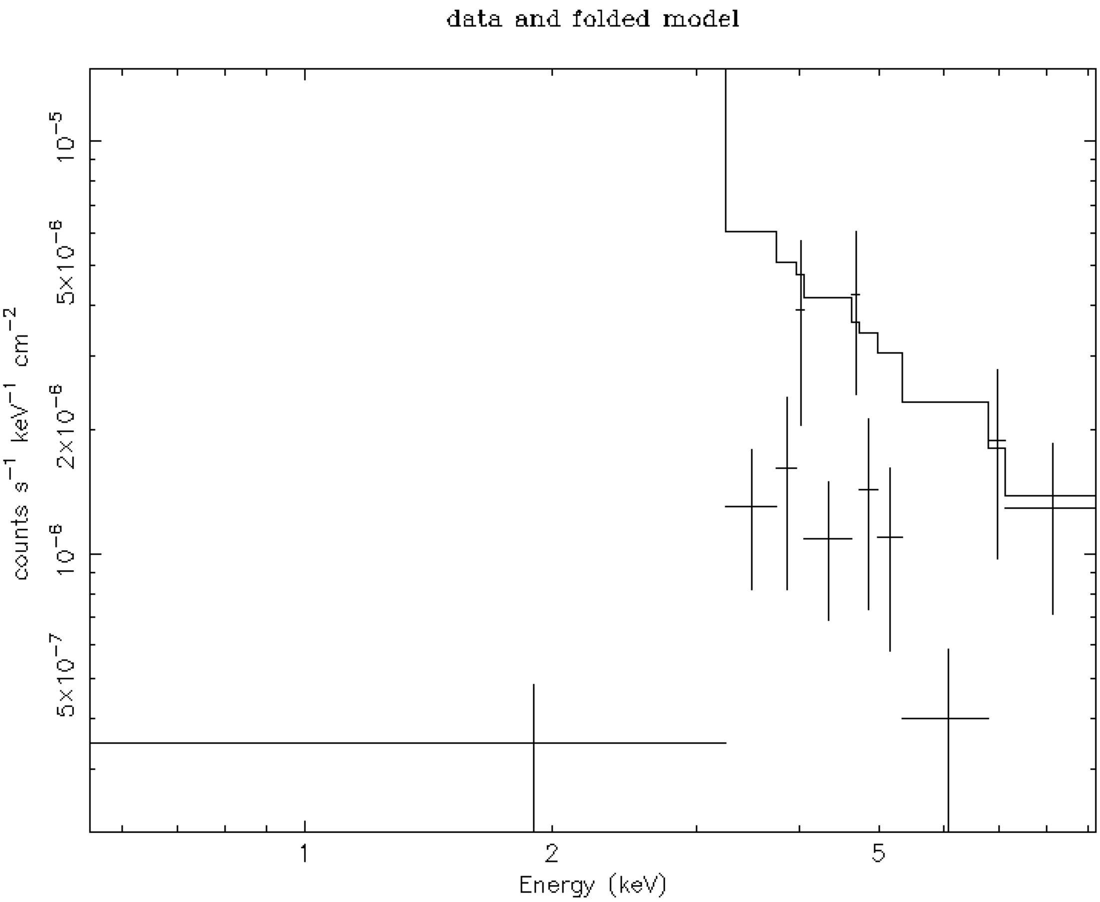

Figure 6: X-ray spectrum of the studied AGN fitted with an absorbed power-law model. The data points are shown with error bars and the best-fit for this model is represented by the line.

The fit is slightly better than the simple power-law model, however it still does not capture all the features of the spectrum.

The corner plot for the absorbed power-law model is shown in Fig. 7.

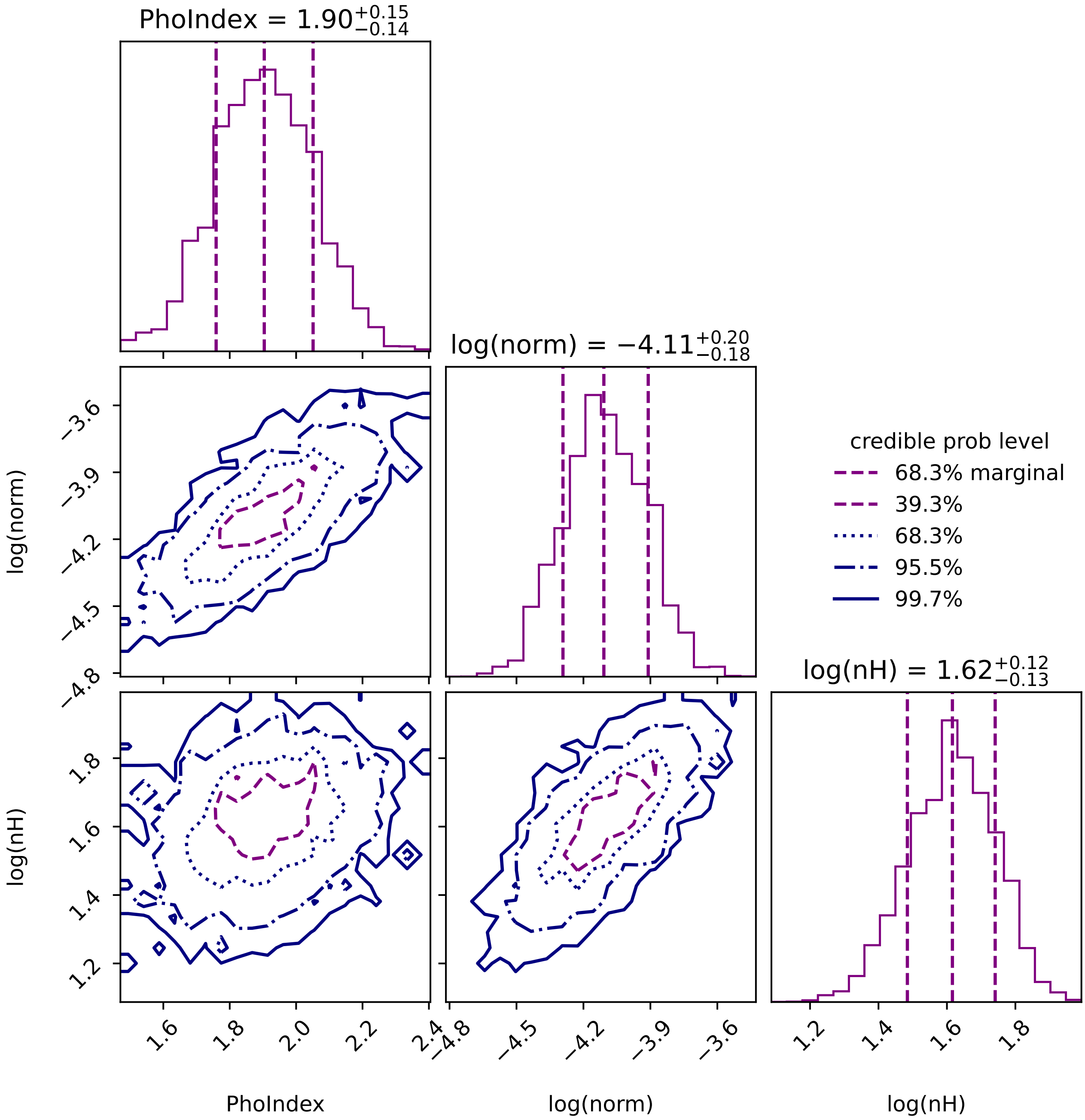

Figure 7: Corner plot of the fitted parameters for the absorbed power-law model. The histograms along the diagonal represent the posterior distributions of each parameter, while the off-diagonal plots show the joint distributions between pairs of parameters, indicating correlations and covariances.

The third and fourth models add more complexity to the absorbed power-law model to better fit the data. As they are not specifically required in the report, the data and model fit plots and the corner plots are not shown here for brevity, but can be found in Appendix B.

The last model is the doble absorbed power-law, which is shown in Fig. 8.

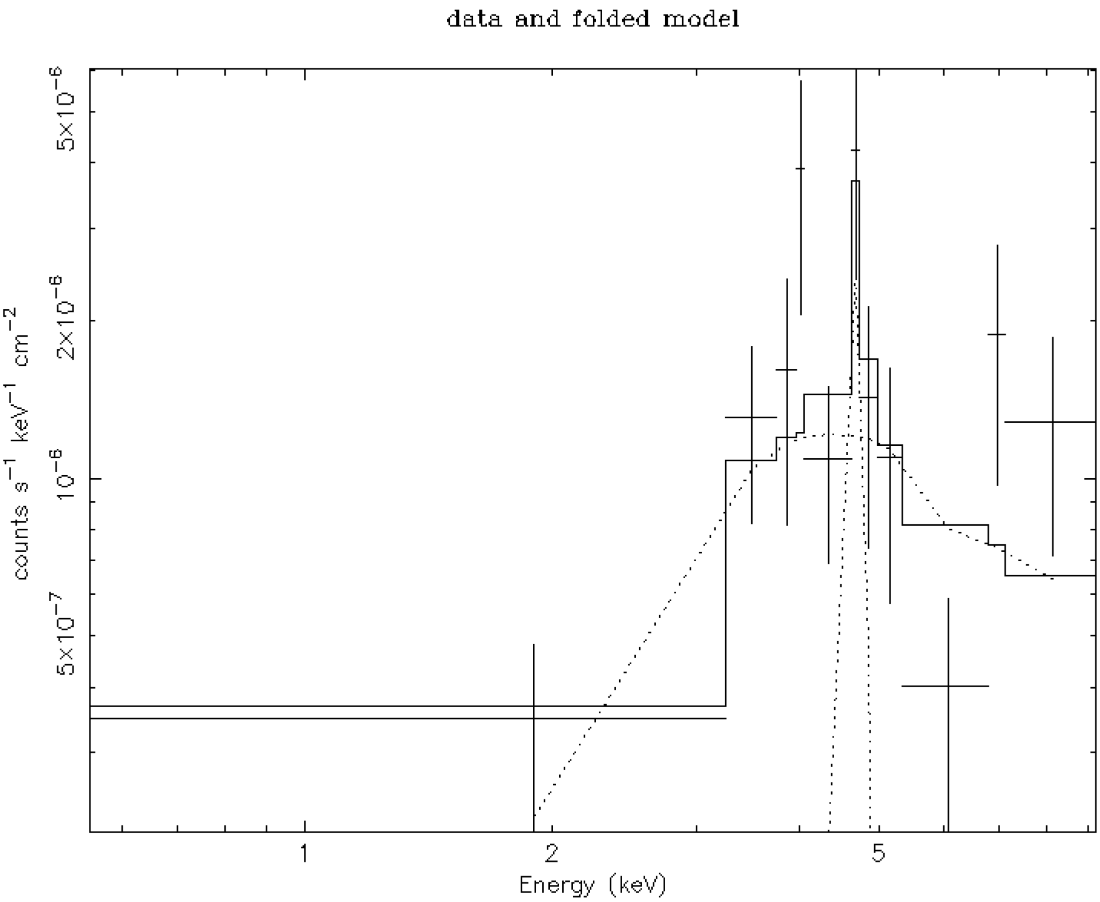

Figure 8: X-ray spectrum of the studied AGN fitted with a doble absorbed power-law model. The data points are shown in black, while the best-fit model is represented by the line.

From the plot in Fig. 8, it can be appreciated that this model fits the data very well, especially in the low-energy part of the spectrum where absorption effects are more pronounced.

The corner plot for the doble absorbed power-law model is shown in Fig. 9.

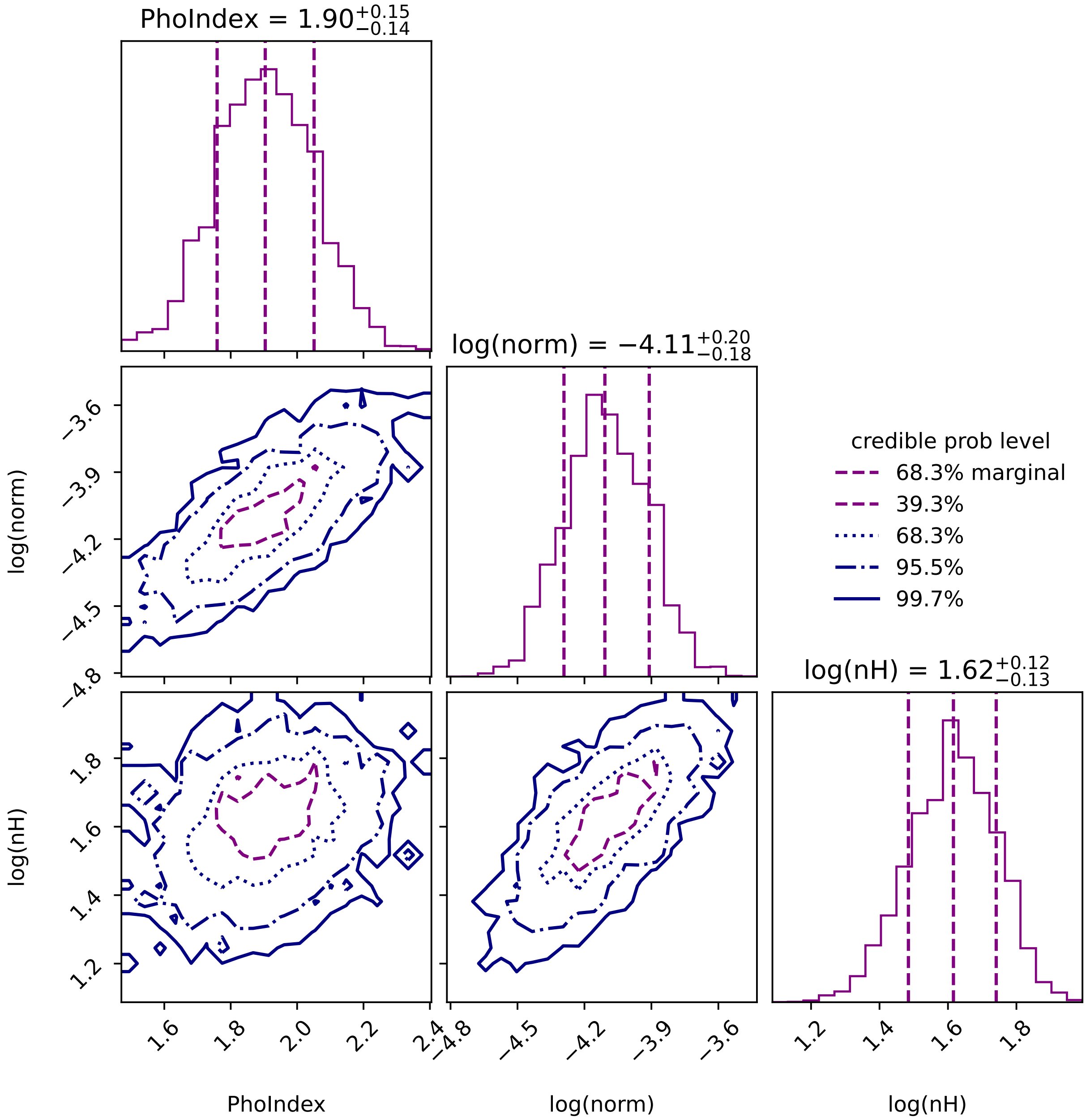

Figure 9: Corner plot of the fitted parameters for the doble absorbed power-law model. The histograms along the diagonal represent the posterior distributions of each parameter, while the off-diagonal plots show the joint distributions between pairs of parameters, indicating correlations and covariances.

The main bayesian statistics results and the AIC values for all the models are shown in Tab. 4.

Model	logZ	PhoIndex	log(norm)	log(nH)	AIC
1. Simple powerlaw	-57.23 ± 0.18	1.03 ± 0.03	-5.63 ± 0.07	–	101.55
2. Abs. powerlaw	-20.60 ± 0.20	1.91 ± 0.15	-4.10 ± 0.18	1.61 ± 0.13	33.34
3. Abs. powerlaw + Gaussian	-42.8 ± 0.4	1.95 ± 0.14	-3.80 ± 0.14	1.97 ± 0.03	72.05
4. Abs. pl + Gaus. + Compton Scat.	-31.6 ± 0.3	1.96 ± 0.15	-3.50 ± 0.17	1.95 ± 0.04	50.53
5. Double Abs. powerlaw	-20.50 ± 0.25	1.7 ± 0.3	-4.18 ± 0.33	1.58 ± 0.14	33.05

Table 4: Bayesian statistics results and AIC values for all the models fitted.

The results in Tab. 4 show that the absorbed power-law (Model 2) and the double absorbed power-law (Model 5) have the highest logZ values, indicating that they are the most likely models given the data. To compare the models more rigorously, the logZ evidence criteria is used, obtaining the following results from the model_compare.py script:

```
(henv) joanribot@Joans-MacBook-Air Source2 \% python3 model\_compare.py
    ↪ powerlaw\_pyxspec/ simplest/ tbabspowerlaw\_pyxspec/ tbabspowerlaw+
    ↪ gaussian\_pyxspec/ tbabspowerlaw+gaussian+cabs\_pyxspec/ double\
    ↪ _powerlaw\_pyxspec/

Model comparison
****************

model simplest/ : log10(Z) = -16.7   XXX ruled out
model powerlaw\_pyxspec/: log10(Z) = -16.0   XXX ruled out
model tbabspowerlaw+gaussian\_pyxspec/: log10(Z) = -9.8   XXX ruled out
model tbabspowerlaw+gaussian+cabs\_pyxspec/: log10(Z) = -4.8   XXX ruled out
model tbabspowerlaw\_pyxspec/: log10(Z) = -0.1    <-- GOOD
model double_powerlaw\_pyxspec/: log10(Z) = 0.0    <-- GOOD
The last, most likely model was used as normalization.
Uniform model priors are assumed, with a cut of log10(30) to rule out models.
```

There are high differences in the two calculated logZ values between the models, as seen in the output above. However, both values conclude that the best models are the absorbed power-law (Model 2) and the double absorbed power-law (Model 5). The AIC results from Tab. 4 confirm the findings from the logZ evidence criteria. The difference between these statistical analyses of the two models is very small, so both can be considered good fits to the data. Both models indicate significant absorption in the X-ray spectrum, consistent with the characteristics of a Type 2 AGN.

2.3 Conclusion and comparison of optical and X-ray results

During the analysis of both the optical and X-ray spectra of the studied AGN some comments have been made about the consistency between the results obtained from both spectra. The main point to highlight is that both analyses suggest that the AGN is likely a Type 2 AGN.

In the optical spectrum, the presence of narrow emission lines and the absence of broad emission lines, such as the Hβ line, indicate that the BLR is obscured, consistent with the characteristics of a Type 2 AGN. However, the narrow line region (NLR) is visible, producing the narrow emission lines observed in the optical spectrum in Fig. 1.

In the X-ray spectrum, the best-fit models indicate significant absorption ($n_H > 10^{22}$ cm^{-2}), which is also characteristic of Type 2 AGNs. Both a power-law with absorption and double absorbed power-law models provide the best fits to the data, supporting the conclusion that the AGN is heavily obscured. All these comments support the classification of the AGN as Type 2 according to the Unified Model.

References

[1] Boorman, P. 2021, *BXA Tutorial: Bayesian X-ray Analysis with UltraNest*, online resource. `https://peterboorman.com/tutorial_bxa.html`

[2] Buchner, J. 2021, *BXA: Xspec Spectral Analysis with Bayesian Statistics*, online resource. `https://johannesbuchner.github.io/BXA/xspec-analysis.html`

[3] Python Software Foundation, *Python Language Reference, version 3.x*, `https://www.python.org`

[4] Arnaud, K. A. 1996, Astronomical Data Analysis Software and Systems V, 101, 17. `https://heasarc.gsfc.nasa.gov/docs/software/xspec/`

[5] Astropy Collaboration et al., *The Astropy Project*, `https://www.astropy.org`

[6] SDSS Collaboration 2024, *SDSS-V DR19 BHM Classroom Tutorial*, Jupyter Notebook, online resource. `https://github.com/sdss/dr19_tutorials/blob/main/BHM_classroom.ipynb`

[7] XQuartz Project Team 2020, *XQuartz: The X Window System on macOS*, online resource. `https://www.xquartz.org/`

[8] Wright, E. L. 2006, PASP, 118, 1711. doi:10.1086/510102 `https://www.astro.ucla.edu/~wright/CosmoCalc.html`

A Run and trace plots for all X-ray models

A.1 X-ray spectrum run and trace plots for simple power-law (Model 1)

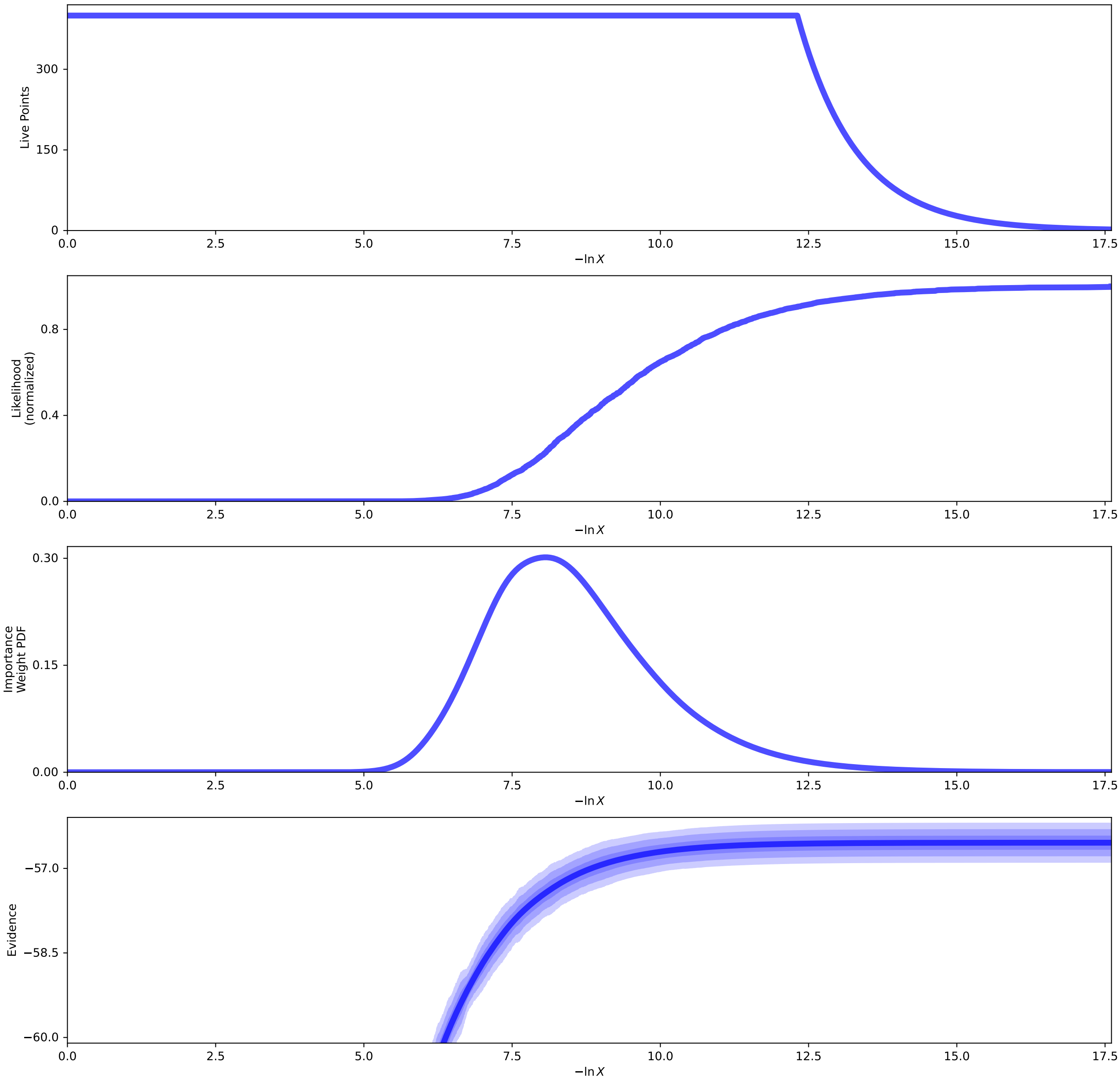

Figure 10: Run plot for the simple power-law model fitted to the X-ray spectrum of the studied AGN.

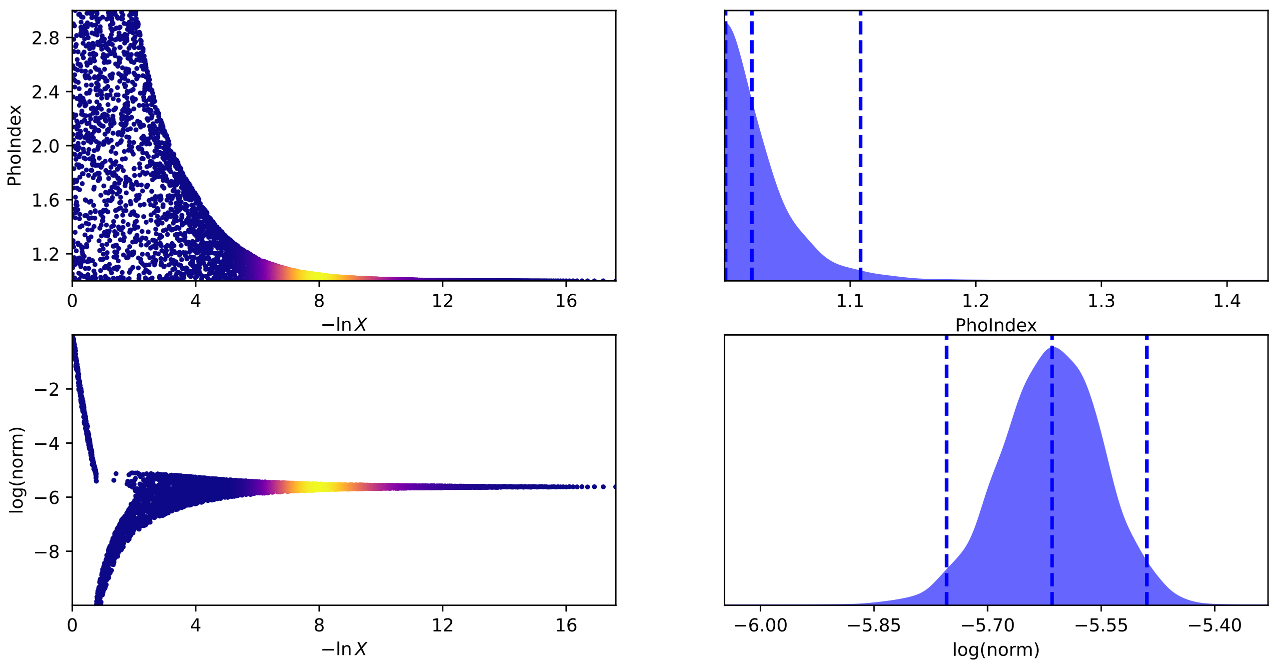

Figure 11: Trace plot for the simple power-law model fitted to the X-ray spectrum of the studied AGN.

A.2 X-ray spectrum run and trace plots for absorbed power-law (Model 2)

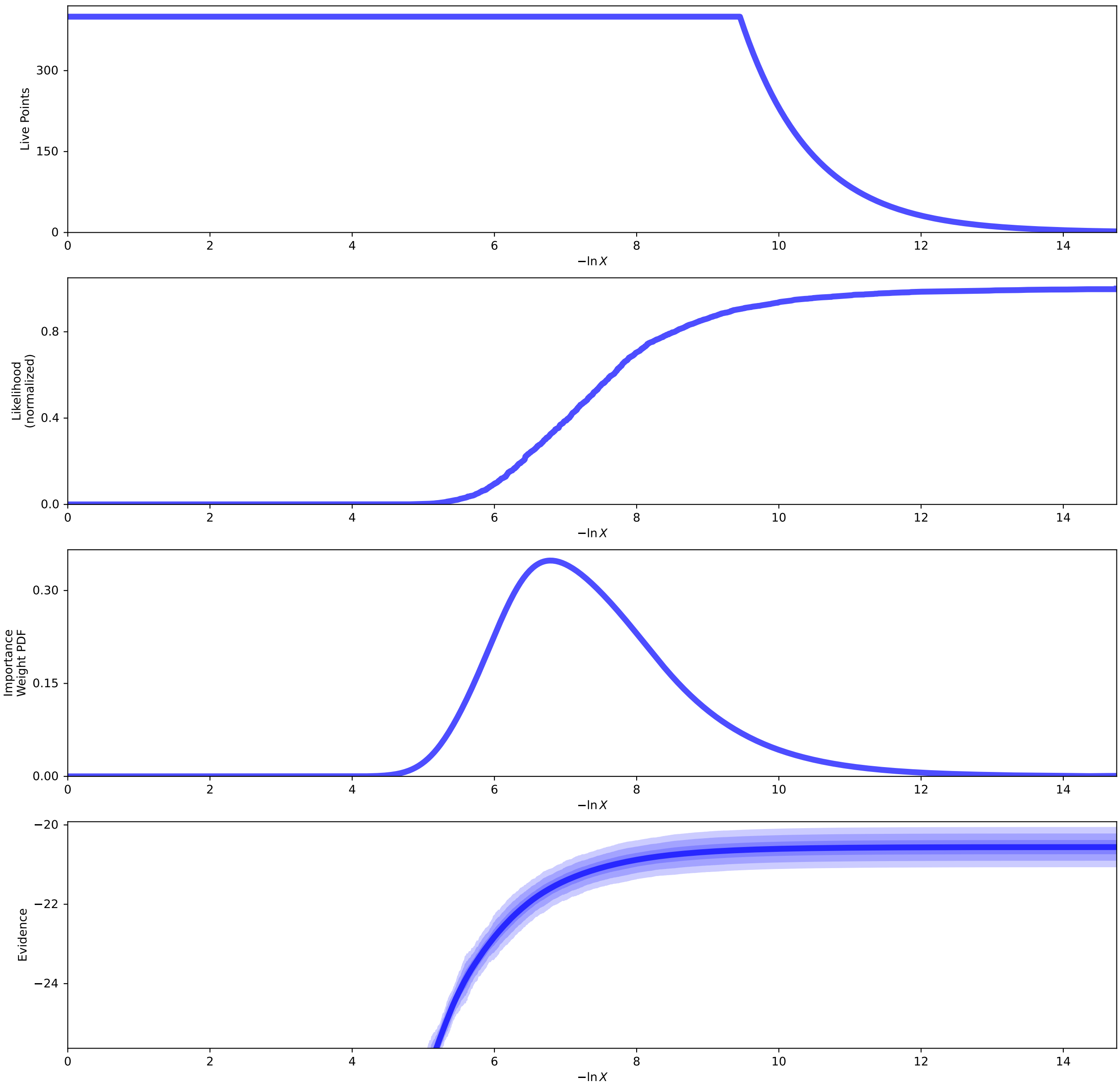

Figure 12: Run plot for the absorbed power-law model fitted to the X-ray spectrum of the studied AGN.

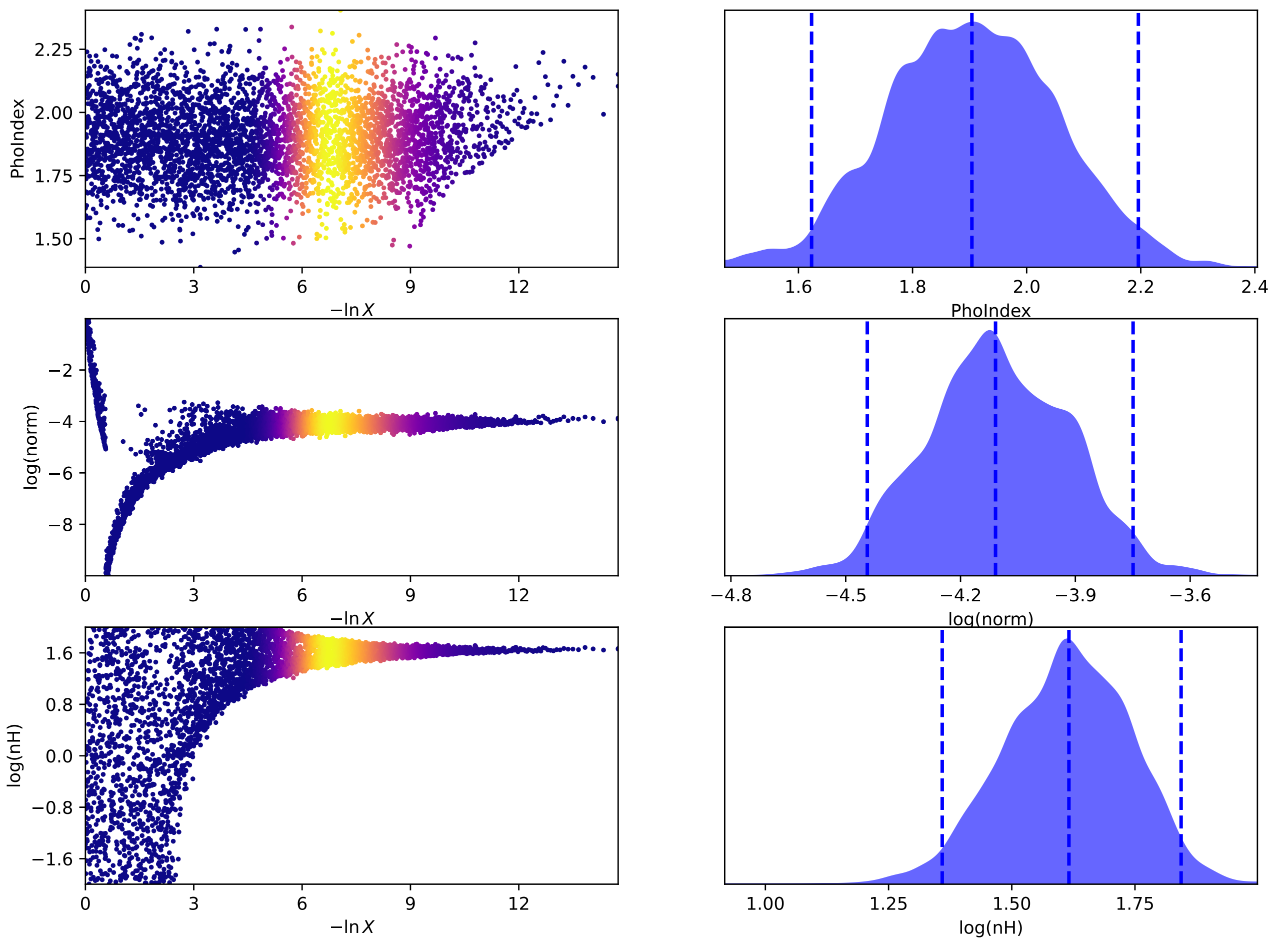

Figure 13: Trace plot for the absorbed power-law model fitted to the X-ray spectrum of the studied AGN.

A.3 X-ray spectrum run and trace plots for absorbed power-law with a Gaussian component (Model 3)

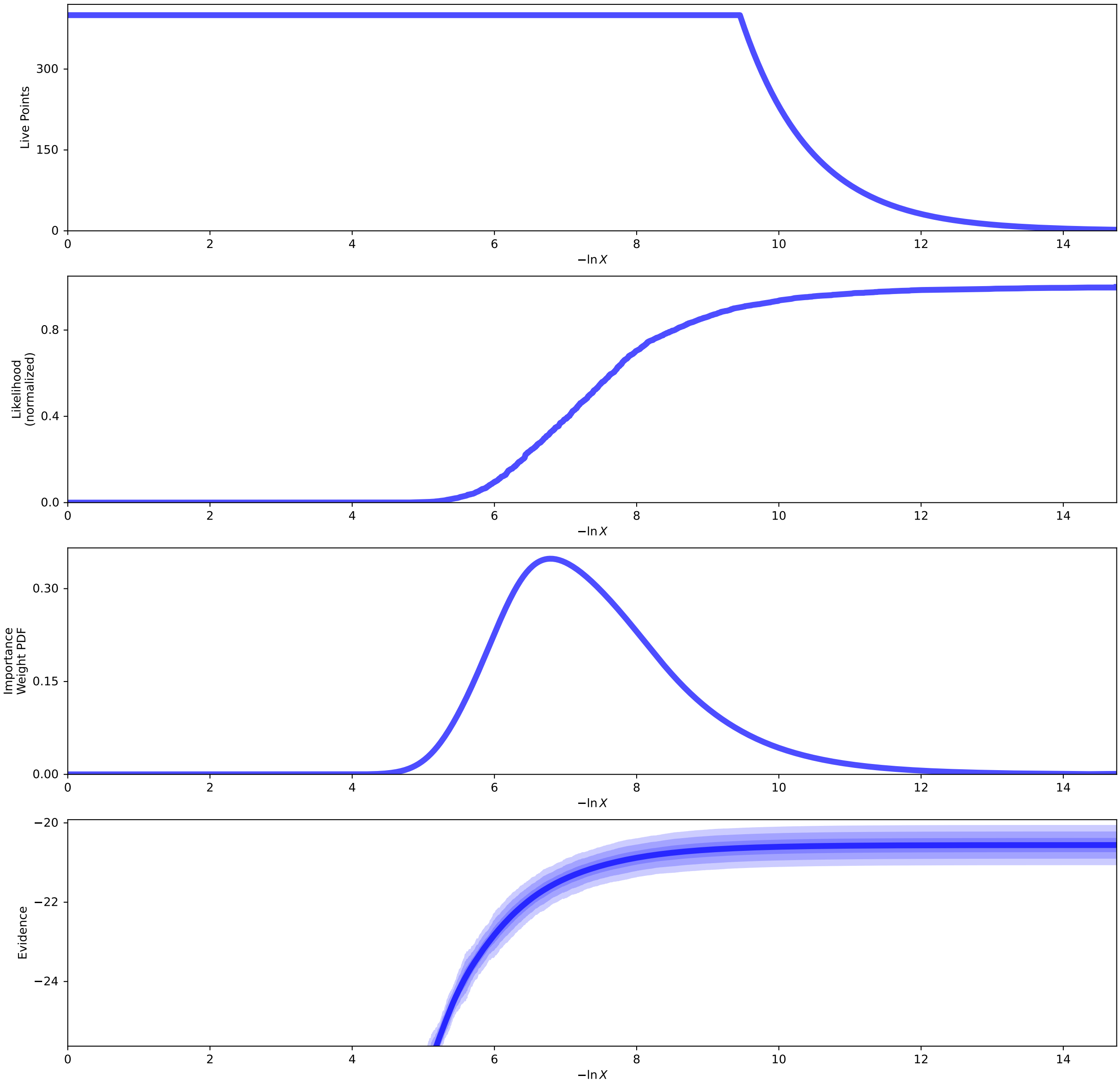

Figure 14: Run plot for the absorbed power-law model fitted to the X-ray spectrum of the studied AGN.

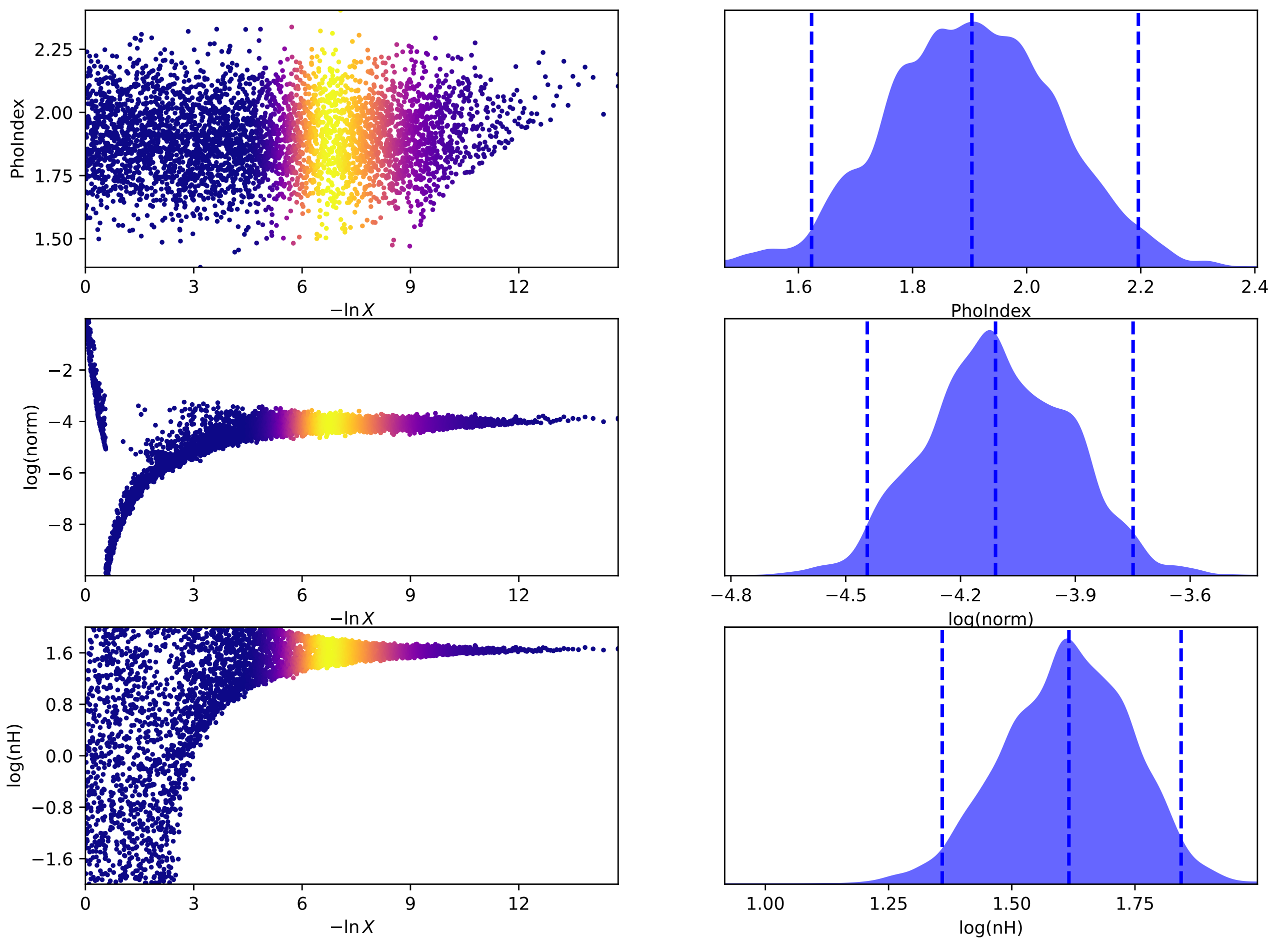

Figure 15: Trace plot for the absorbed power-law model fitted to the X-ray spectrum of the studied AGN.

A.4 X-ray spectrum run and trace plots for absorbed power-law with a Gaussian component and Compton scattering (Model 4)

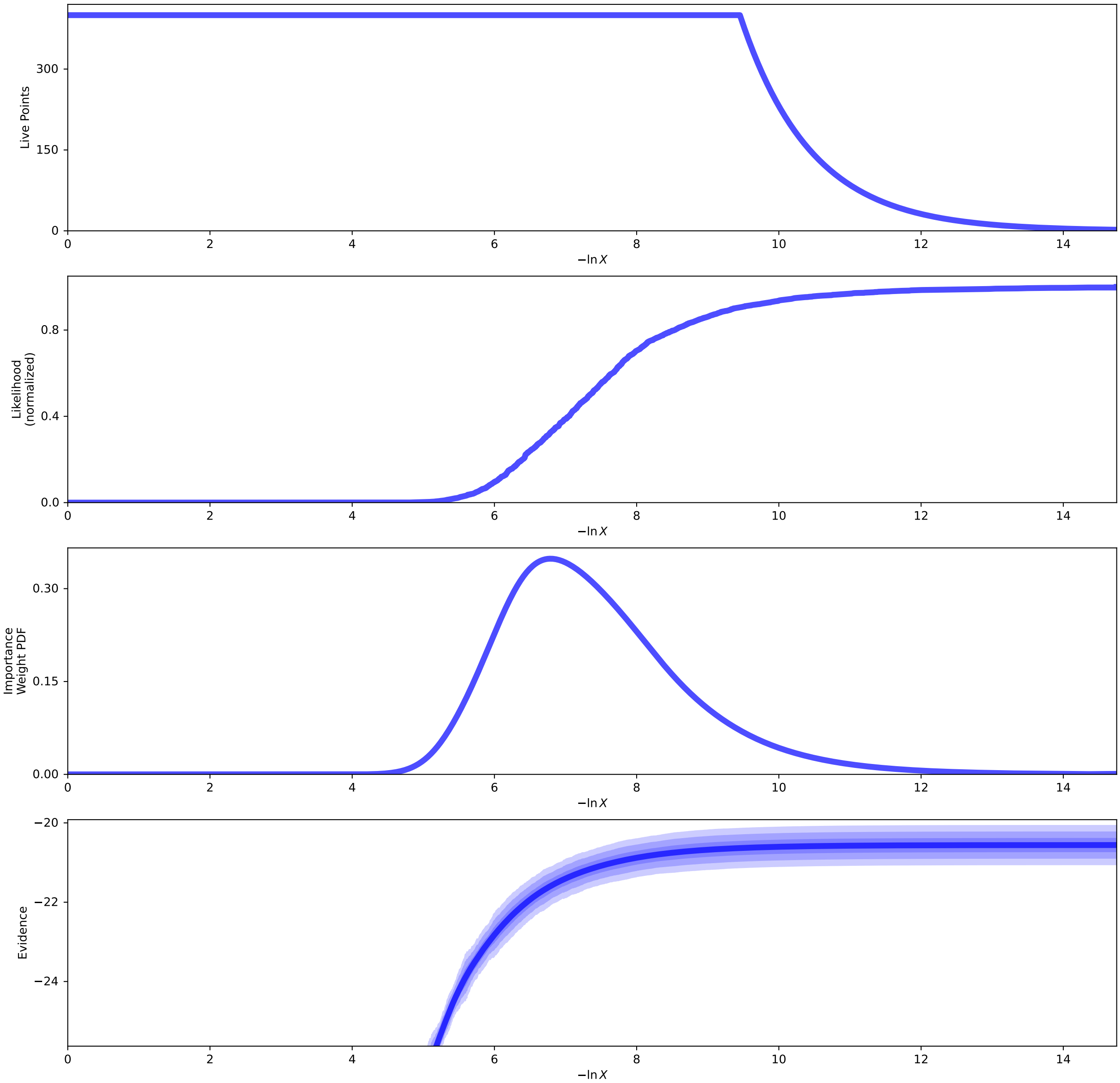

Figure 16: Run plot for the absorbed power-law model fitted to the X-ray spectrum of the studied AGN.

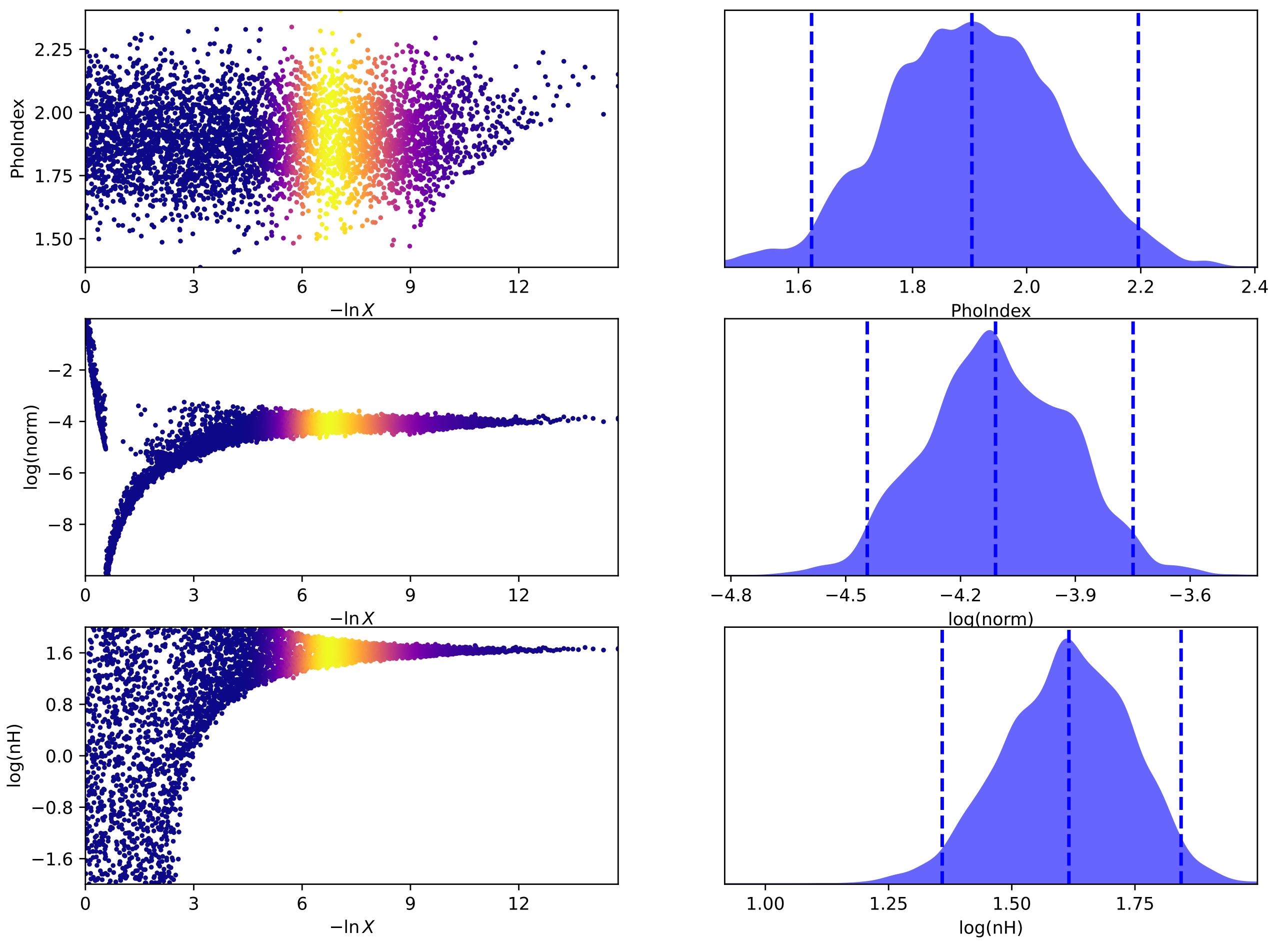

Figure 17: Trace plot for the absorbed power-law model fitted to the X-ray spectrum of the studied AGN.

A.5 X-ray spectrum run and trace plots for doble absorbed power-law (Model 5)

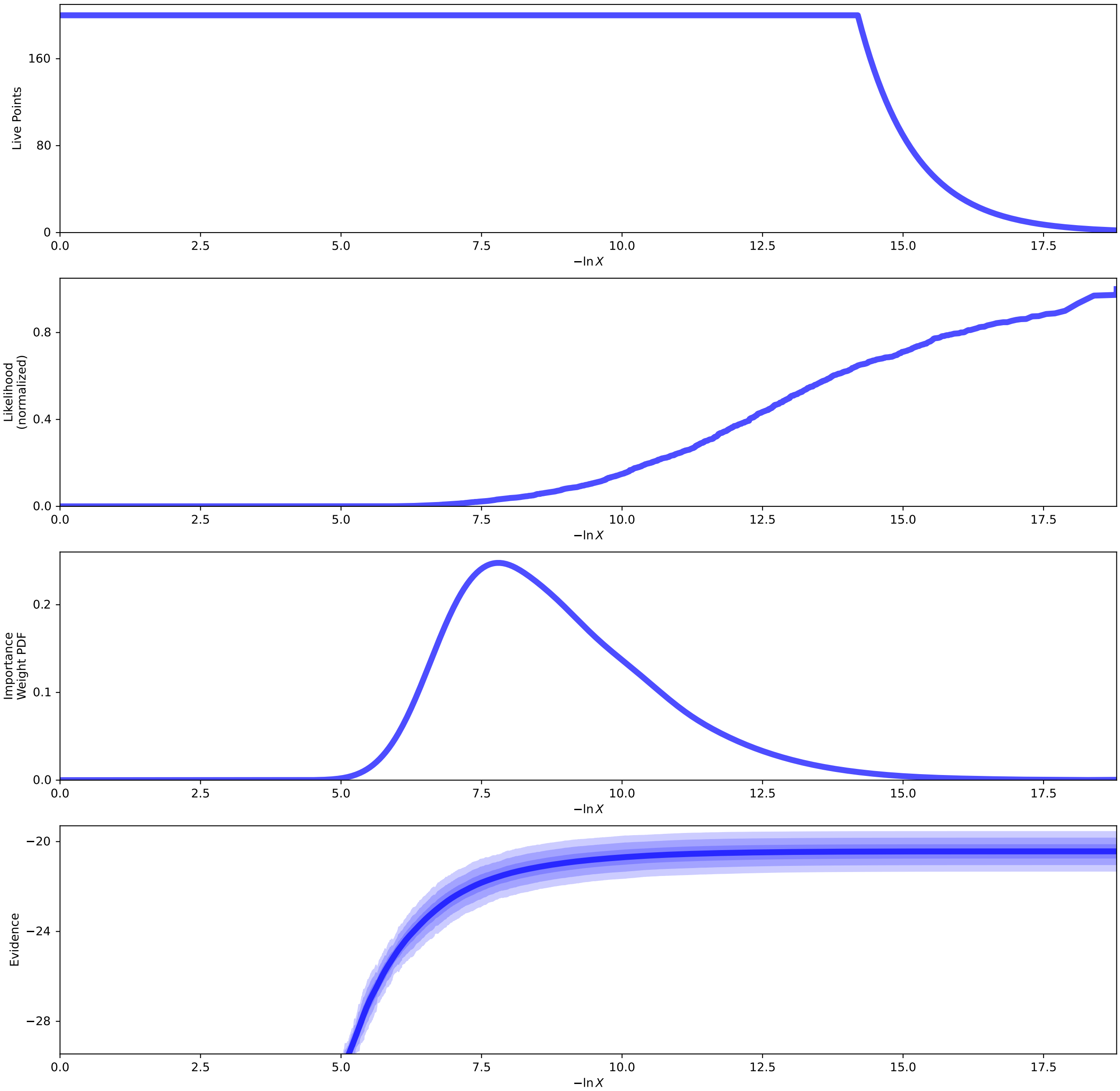

Figure 18: Run plot for the doble absorbed power-law model fitted to the X-ray spectrum of the studied AGN.

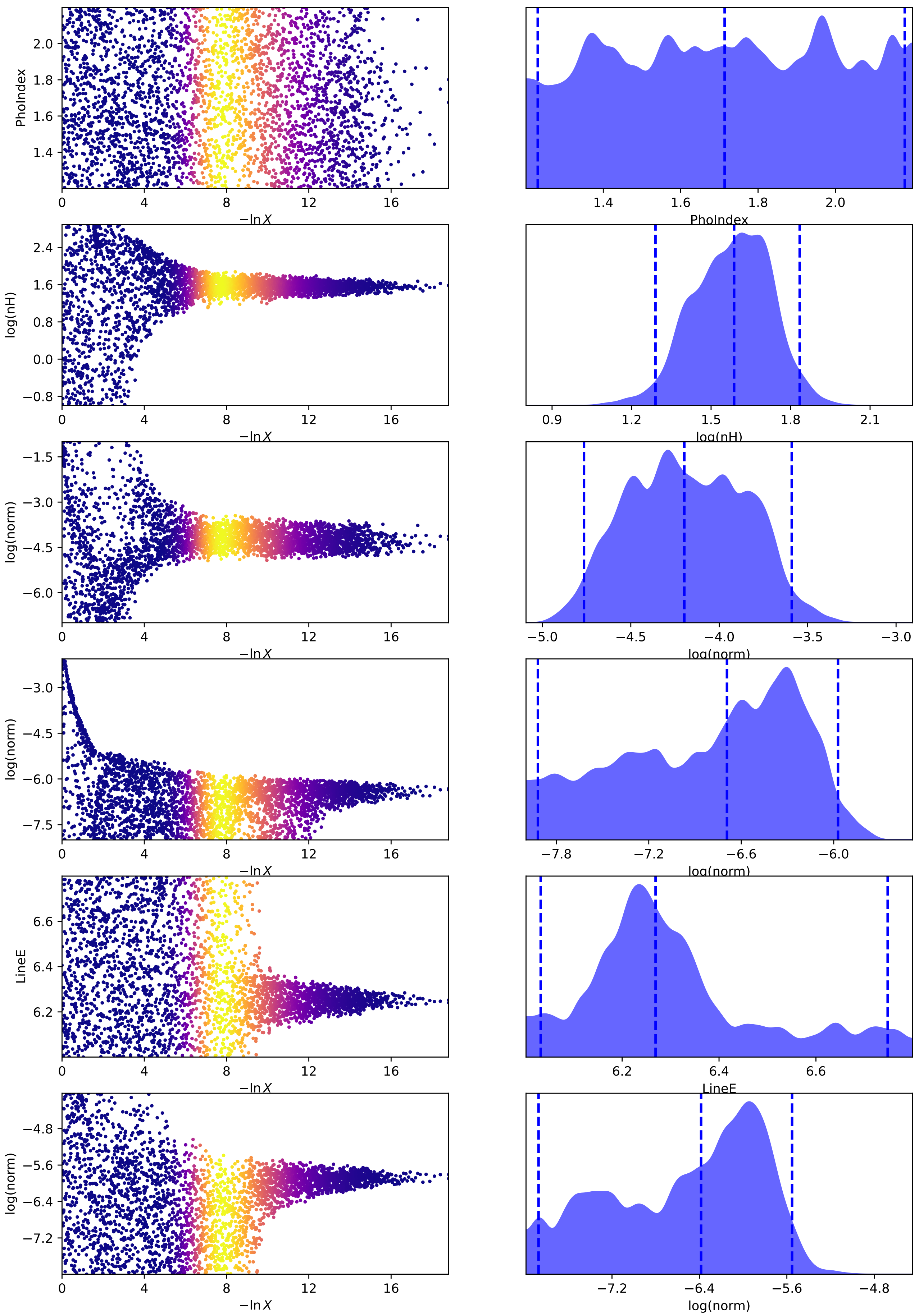

Figure 19: Trace plot for the doble absorbed power-law model fitted to the X-ray spectrum of the studied AGN.

B Data and model plots and corner plots for models 3 and 4

B.1 X-ray spectrum data and model plots and corner plots for absorbed power-law with a Gaussian component (Model 3)

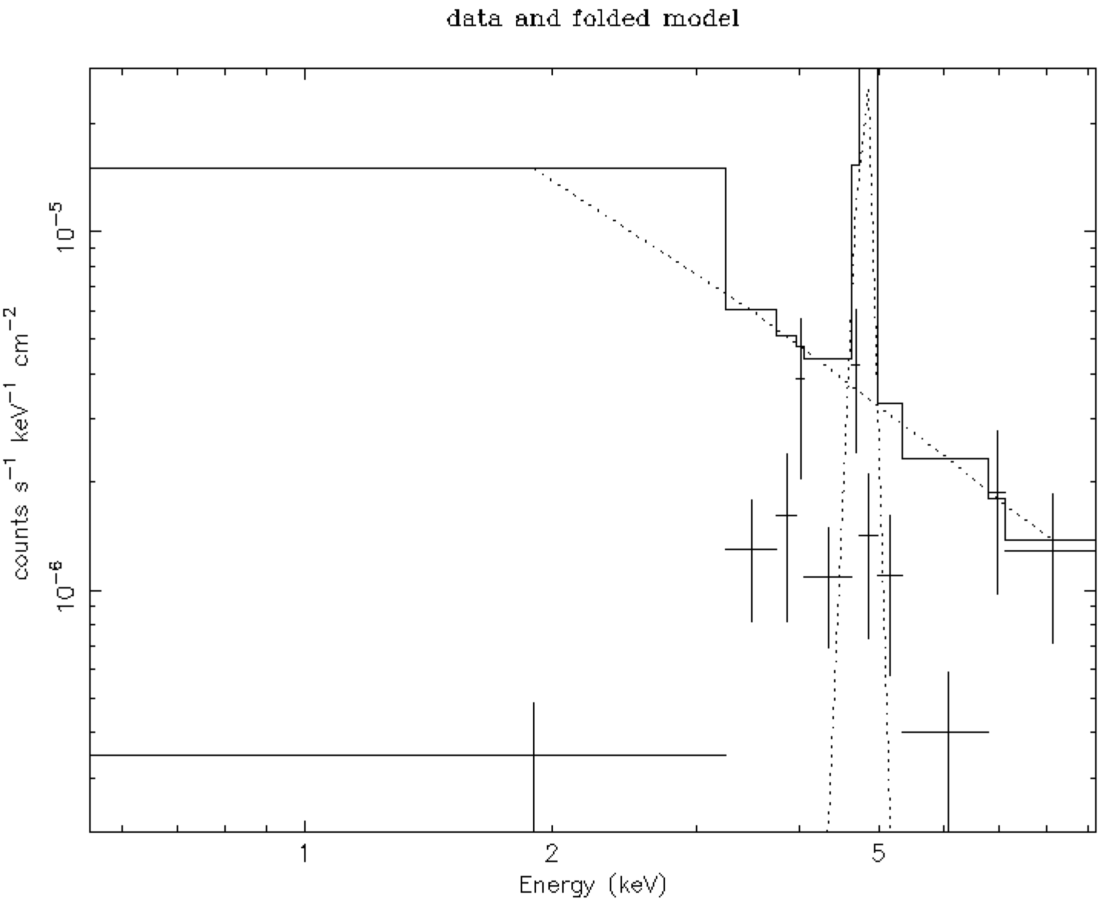

Figure 20: X-ray spectrum of the studied AGN fitted with an absorbed power-law with a Gaussian component model. The data points are shown with error bars and the best-fit model is represented by the line.

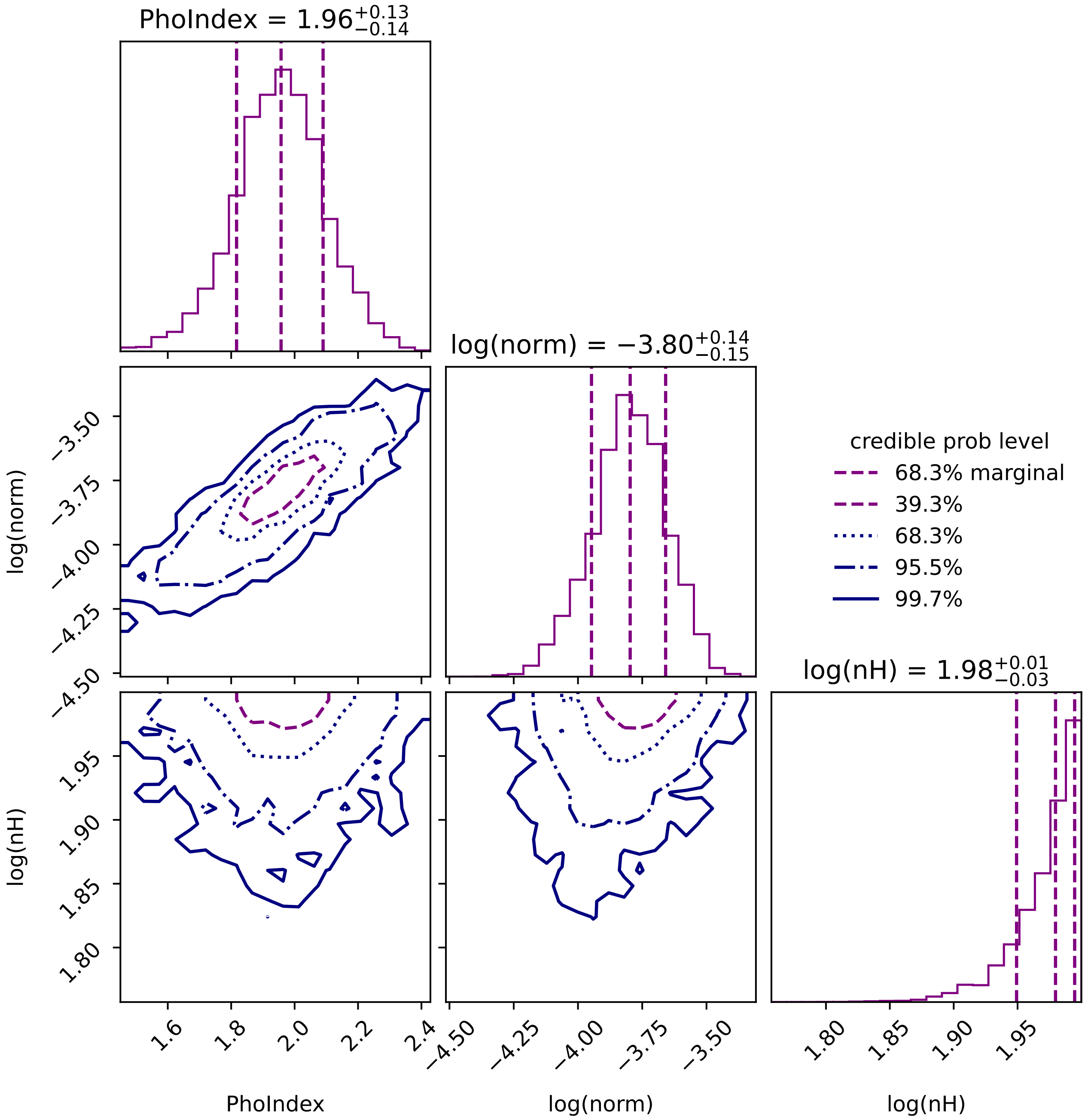

Figure 21: Corner plot of the fitted parameters for the absorbed power-law with a Gaussian component model. The histograms along the diagonal represent the posterior distributions of each parameter, while the off-diagonal plots show the joint distributions between pairs of parameters, indicating correlations and covariances.

B.2 X-ray spectrum data and model plots and corner plots for absorbed power-law with a Gaussian component and Compton scattering (Model 4)

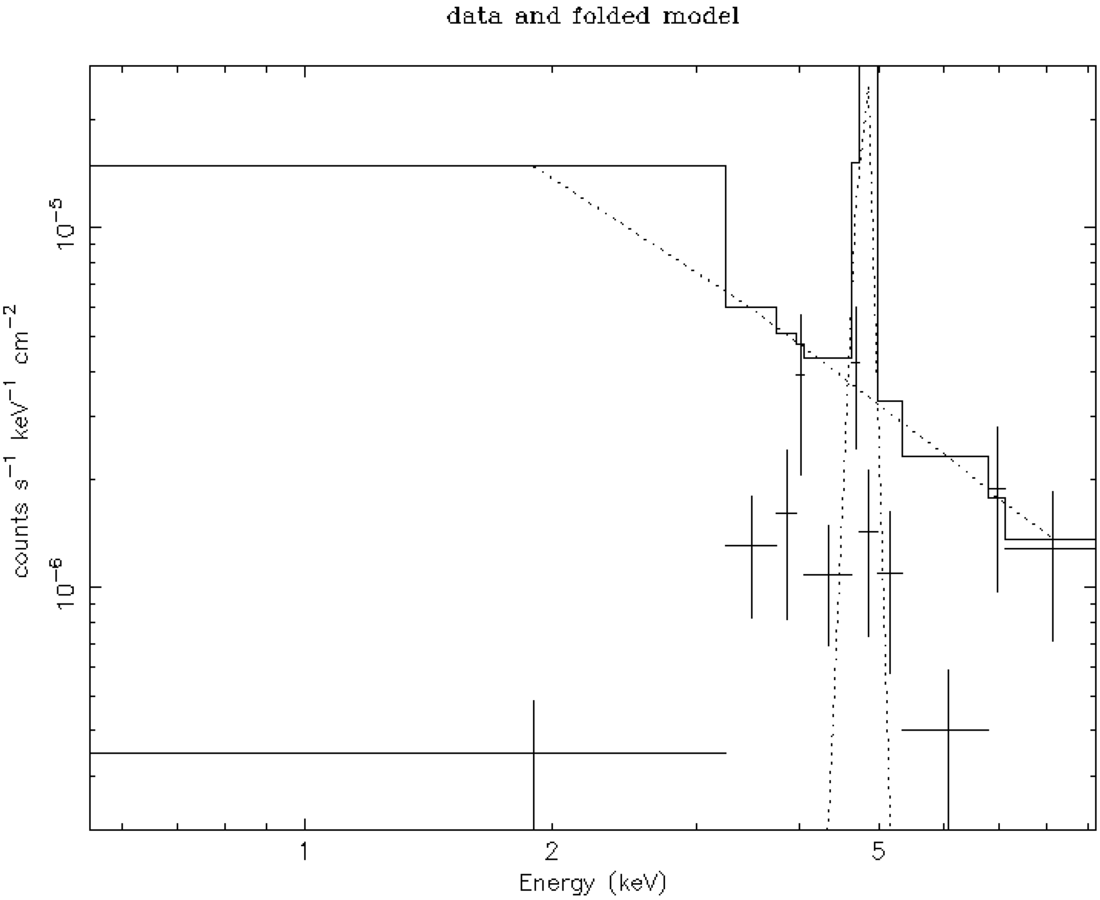

Figure 22: X-ray spectrum of the studied AGN fitted with an absorbed power-law with a Gaussian component and Compton scattering model. The data points are shown with error bars and the best-fit model is represented by the line.

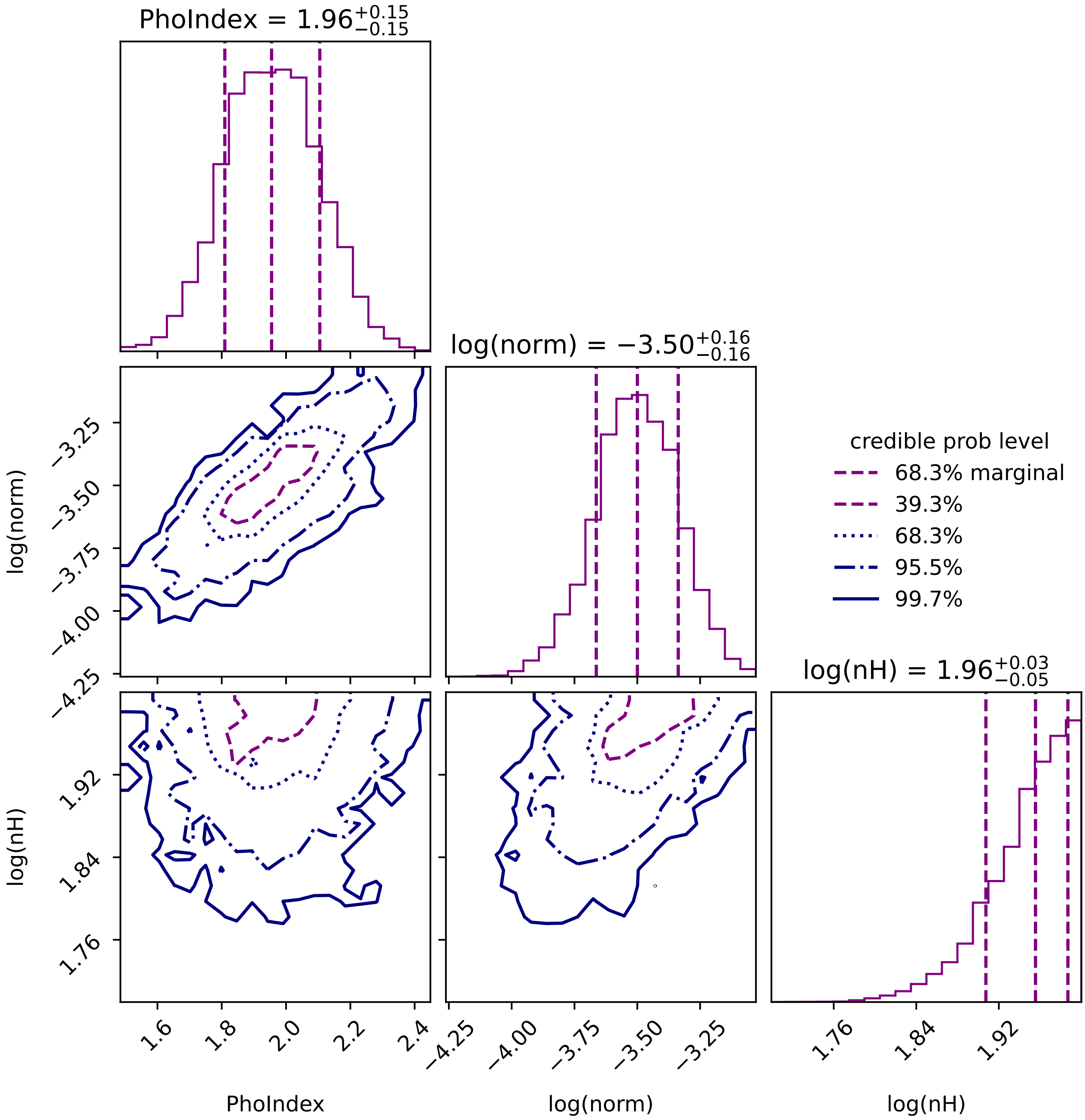

Figure 23: Corner plot of the fitted parameters for the absorbed power-law with a Gaussian component and Compton scattering model. The histograms along the diagonal represent the posterior distributions of each parameter, while the off-diagonal plots show the joint distributions between pairs of parameters, indicating correlations and covariances.